# Lighting & Phong Reflection Model – Lecture Notes

■ 1. FUNDAMENTALS OF LIGHTING
- Illumination: Energy transport from light sources to surfaces.
• Local Illumination – Depends on direct light.
• Global Illumination – Includes reflection/refraction.
- Lighting Model: Determines surface color or brightness based on lighting.

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■ 2. TYPES OF LIGHT SOURCES
- Ambient Light: No direction; background light.
- Diffuse Light: Scattered reflection; follows Lambert's cosine law.
- Specular Light: Mirror-like reflection; depends on viewer.

Formula: Total Illumination = Ambient + Diffuse + Specular

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■ 3. PHONG REFLECTION MODEL
I = Iaka + Ip [kd max(L·N, 0) + ks max(R·V, 0)^n]

Where:
• I = total light intensity at point
• Ia = ambient intensity, Ip = point light intensity
• ka, kd, ks = ambient, diffuse, specular coefficients
• L = Light vector, N = Normal, R = Reflected ray, V = View vector
• n = shininess exponent

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■ 4. COMPONENTS EXPLAINED

• Ambient Reflection:
- Uniform light, simulates global bounce.
- Not viewer or angle dependent.
- Iamb = Iaka

• Diffuse Reflection (Lambertian):
- Based on angle between L and N
- Idiff = Ip * kd * max(0, L · N)

• Specular Reflection:
- Depends on angle between R and V
- I_spec = Ip * ks * max(0, R · V)^n

Blinn-Phong variation:
Replace R·V with N·H where H = half vector between L and V.

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■ 5. SPOTLIGHT
- Emits light in a conical region.
- Requires direction, cutoff angle θ, and falloff function.
- Intensity drops outside cutoff.